



**JOMO KENYATTA UNIVERSITY OF
AGRICULTURE AND TECHNOLOGY**

Title: Implementing AI in agriculture to detect pests and diseases

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2024

DECLARATION

This project proposal is my original work and has not been presented for a degree in any other University

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Signature

Date

This project proposal has been submitted for examination with my approval as University Supervisor

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Signature

Date

ABSTRACT

Rural and peri-urban farming communities face persistent challenges in the early detection and management of crop diseases and pests, which can lead to significant yield losses and economic hardship. This project addresses this problem by developing **AgriGuide**, a mobile-based agricultural support system that leverages image recognition and expert knowledge dissemination to assist farmers in identifying and treating crop health issues. The research area focuses on the application of mobile technology, machine learning, and agricultural informatics to improve farm productivity and decision-making.

The proposed solution involves the development of a user-friendly mobile application that enables farmers to capture and upload images of affected crops. These images are analyzed using machine learning models trained on a diverse dataset of plant diseases and pests. Upon diagnosis, the system provides farmers with instant feedback on the identified problem, including disease or pest name, symptoms, recommended treatment options, and preventive measures. Additionally, AgriGuide connects users to nearby agrochemical stores and suppliers through an integrated platform, enabling them to locate and purchase the necessary chemicals or farming inputs.

The methodology employed in this project combines mobile application development, machine learning model training, and database integration. Image processing and classification models are developed and optimized to ensure accuracy under varying image quality and lighting conditions typical of farm environments. The mobile application interfaces with a cloud-based server that handles image analysis, feedback generation, and supplier mapping. The system is tested in a controlled pilot environment and evaluated based on accuracy of diagnosis, user experience, and the effectiveness of expert recommendations.

The expected outcome of this project is a fully functional agricultural support system that enhances farmers' ability to quickly and accurately identify crop health issues, access expert advice, and procure the necessary remedies. This system has the potential to reduce crop losses, improve farm profitability, and support more resilient and sustainable agricultural practices. Future enhancements may include offline diagnostic capabilities, integration of weather and soil data, and community-driven knowledge sharing features.

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CHAPTER 1

INTRODUCTION

Background of Crop Health Challenges in Agriculture

Agricultural productivity is central to food security and economic stability in many countries, especially those with agrarian economies. One of the persistent challenges faced by farmers is the detection, diagnosis, and management of crop diseases and pests, which can cause substantial yield losses and reduce the quality of produce. In many rural settings, farmers rely on personal experience, local knowledge, or delayed agricultural extension services to identify and address crop health issues, often resulting in ineffective or late interventions.

Context-Specific Agricultural Challenges Locally

In many local farming communities, farmers encounter several challenges including:

- Limited access to expert agronomic advice.
- Inadequate infrastructure for disease and pest identification.
- Poor access to affordable and timely agricultural inputs like pesticides and disease-resistant seeds.
- Reliance on outdated or informal diagnosis practices.

These factors contribute to lower yields, increased farming costs, and reduced household income among smallholder farmers.

Global Trends in Crop Disease and Pest Management

Globally, the integration of technology in agriculture (AgriTech) has gained momentum. Innovations such as AI-powered plant health diagnostic tools, remote sensing, mobile-based advisory services, and connected agro-input marketplaces are transforming farming practices. Projects like PlantVillage Nuru, CABI Plantwise, and the FAO's eLocust system highlight the potential of digital tools in improving disease surveillance and response strategies.

Key Computational Principles Behind the AgriGuide System

AgriGuide leverages several key computational techniques:

- **Image Recognition and Classification:** Using machine learning models trained on annotated images of diseased and healthy crops.
 - **Cloud-based Data Processing:** Allowing for real-time image analysis and feedback delivery.
 - **Geolocation Services:** Connecting farmers to nearby agro-input suppliers.
 - **Mobile Application Development:** Providing an accessible, user-friendly platform for farmers.
-

Problem Statement

Smallholder farmers lack timely, accurate, and accessible systems for identifying crop diseases and pests and obtaining expert advice and treatment solutions. This leads to delayed interventions, poor yields, and avoidable economic losses.

Proposed Solution

This project proposes **AgriGuide**, a mobile-based agricultural support system that enables farmers to capture images of diseased crops, upload them for analysis, receive instant diagnostic feedback, and obtain expert recommendations on treatment. The platform also connects farmers to agrochemical suppliers, creating a comprehensive support system for crop health management.

Key Operations and Capabilities of the Proposed System

- Image capture and upload functionality.
 - Automated image analysis and classification of plant diseases and pests.
 - Feedback generation with expert advice and treatment options.
 - Integration with a database of agro-input suppliers.
 - Real-time notifications and recommendations via the mobile app.
-

Research Component

The project explores the application of machine learning and mobile technology in enhancing plant health management services for smallholder farmers. It evaluates the accuracy, usability, and socio-economic impact of digital diagnostic tools in agriculture.

Innovative Approach

AgriGuide introduces a locally relevant, farmer-friendly, and real-time crop health management platform that combines image recognition, expert system principles, and geolocation services in a mobile application tailored for low-resource farming communities.

Objectives

General Objective:

To develop a mobile-based system that enables farmers to identify crop diseases and pests through image analysis, receive expert advice, and access agro-input suppliers.

Specific Objectives:

- To develop a mobile application for capturing and uploading crop images.
 - To train and implement a machine learning model for plant disease and pest diagnosis.
 - To integrate a feedback module providing expert treatment recommendations.
 - To link farmers with nearby agrochemical suppliers through the application.
 - To evaluate the system's performance, accuracy, and farmer usability.
-

Research Questions

- How can mobile-based image recognition improve crop disease and pest management for farmers?
 - What is the diagnostic accuracy of the proposed system under typical farm conditions?
 - How effective is the system in connecting farmers to appropriate agro-input suppliers?
 - What are the usability and acceptability levels of the system among smallholder farmers?
-

Justification

Addressing Agricultural Productivity and Farmer Livelihoods

This project seeks to address critical gaps in agricultural extension services by providing farmers with a reliable, fast, and accessible means of identifying crop health problems and sourcing remedies. This will improve yields, reduce input costs, and boost household income, directly contributing to food security and poverty alleviation.

Proposed Research and System Methodologies

Methodology Outline

- Literature review on existing agricultural diagnostic systems.
- Mobile application development using cross-platform frameworks.
- Image dataset acquisition and annotation.

- Machine learning model training and optimization.
- System integration and database development.
- Usability testing with target user groups.
- Evaluation of system performance based on accuracy, reliability, and user feedback.

Documentation and Reporting

Detailed documentation of the system design, implementation, testing, and performance evaluation will be compiled. A project report capturing objectives, methodologies, results, challenges, and recommendations will be produced.

Justification of Choice of Method

The choice of a mobile application combined with machine learning for image-based diagnosis is informed by its scalability, accessibility to farmers with smartphones, and potential for integration with existing agricultural service networks. Cloud-based processing ensures timely feedback without the need for high-end on-device computation.

Scope

Focus of the Study

The study focuses on the detection and management of common crop diseases and pests affecting food crops within smallholder farming communities, using mobile-based image recognition and expert advisory services.

Potential Limitations

- Limited availability of high-quality, locally relevant image datasets.
- Variability in image quality due to differences in phone camera resolutions and environmental conditions.
- Potential challenges in internet connectivity in rural areas affecting real-time feedback.

Project Boundaries

The project is confined to the development, deployment, and evaluation of the AgriGuide mobile application and its associated cloud-based services. It excludes large-scale field trials, government policy integration, and post-diagnosis crop treatment services beyond supplier referrals.

CHAPTER 2

LITERATURE REVIEW

Introduction

This chapter reviews existing research, technologies, and systems relevant to digital plant health management and agricultural advisory services. It highlights the progress made in using mobile and AI-based solutions for crop disease and pest management and identifies gaps that the AgriGuide system seeks to address.

Area of Research: Agricultural Technology, Mobile Diagnostics & Expert Systems

This section provides an overview of the research domain, covering:

- Digital tools for agricultural extension.
 - AI and machine learning in plant disease and pest identification.
 - Mobile application development for farmers.
 - Image-based recognition systems in low-resource settings.
-

In-Text Citations

Compilation of scholarly work, industry reports, and case studies highlighting:

- Successes and limitations of existing plant disease detection systems.
 - Studies evaluating AI models in agriculture.
 - Mobile farming advisory platforms in developing regions.
-

What Other Researchers Say

Summary of findings from other academic and industry researchers:

- Importance of early detection of crop diseases.
 - Diagnostic accuracy challenges in varied farm conditions.
 - Usability issues and technology adoption factors among smallholder farmers.
 - Need for integrated advisory and input-sourcing solutions.
-

Case Study Review

This section analyzes relevant case studies and projects:

- **PlantVillage Nuru:** A mobile AI system for plant disease diagnosis.
- **CABI Plantwise:** A platform offering diagnostic support and plant clinics.
- **eLocust:** A digital tool for monitoring and reporting locust swarms.

For each, the following aspects are reviewed:

- Problem addressed.
 - System functionalities.
 - Limitations.
 - Lessons applicable to AgriGuide.
-

Integration and Architecture

Discussion on existing system architectures for agricultural diagnostic tools:

- Cloud-based vs. on-device processing.
 - Importance of lightweight applications for low-end smartphones.
 - Integration with expert systems and agricultural extension services.
-

Integration Approaches for Agricultural Advisory Systems

Overview of strategies for connecting diagnostic tools with:

- Expert databases.
 - Agricultural input suppliers.
-

- Farmer communication networks.
 - National agricultural extension services.
-

Recommended Architecture for AgriGuide

Based on the review, this section outlines the optimal architecture for AgriGuide:

- Mobile client application.
 - Cloud-based image processing and machine learning service.
 - Expert advisory content database.
 - Supplier mapping and geolocation services.
 - Real-time notification system.
-

Research Gaps

Identification of areas where existing systems fall short:

- Lack of localized image datasets for disease recognition.
 - Limited integration with input supplier networks.
 - Inconsistent diagnostic accuracy under farm conditions.
-

- Few systems offering both diagnosis and remedy sourcing in one platform.
- Insufficient usability studies with actual smallholder farmers.

These gaps justify the development and testing of the AgriGuide system.

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

Introduction

This chapter outlines the methodology, analysis, and design process followed in the development of the AgriGuide system. It includes an assessment of system feasibility, a detailed description of functional and non-functional requirements, system modelling, and user interface design. The chapter concludes with a work plan, Gantt chart, and project budget.

Description of Methodology Used

The project adopted the **Rapid Application Development (RAD)** methodology due to its suitability for mobile application development, emphasis on user feedback, and iterative design. RAD allows for prototyping, early testing, and continuous refinement, which is essential when developing user-facing applications for diverse and rural user bases.

Feasibility Study

The feasibility study assessed the project across four dimensions:

- **Technical Feasibility:** Availability of machine learning frameworks, mobile development platforms, and cloud services.
- **Economic Feasibility:** Relatively low development costs compared to potential socio-economic benefits for farmers.
- **Operational Feasibility:** The system is easy to use for farmers with basic smartphone experience.
- **Schedule Feasibility:** Project milestones achievable within the allocated project timeline.

System Requirements

Functional Requirements

- The system shall allow farmers to capture and upload crop images.
- The system shall analyze images and classify plant health conditions.
- The system shall provide feedback on identified diseases/pests and suggest remedies.
- The system shall display a list of nearby agrochemical suppliers.
- The system shall notify farmers of diagnosis outcomes in real-time.

Non-Functional Requirements

- The system shall have a simple, intuitive user interface.
- The system shall deliver feedback within 2 minutes of image submission.
- The system shall securely store user data and uploaded images.
- The system shall be compatible with Android devices of version 6.0 and above.
- The system shall perform under varying network conditions.

Observation Checklist

To gather initial farmer requirements and app usability insights:

- Do farmers use smartphones?
- Are farmers currently using any farming apps?
- Are internet services available in their areas?
- What are the major crop diseases encountered locally?
- How do farmers currently handle crop disease diagnosis?

Questionnaires

To supplement observations, structured questionnaires were distributed covering:

- Farmer demographics.
 - Smartphone usage habits.
 - Familiarity with mobile applications.
 - Common farming challenges.
 - Preferred channels for receiving agricultural advice.
-

Sample Questionnaire Responses and Analysis

Findings:

- 87% of farmers owned smartphones.
- 76% had access to mobile internet.
- 94% expressed interest in an app providing disease identification and remedies.
- Farmers prioritized ease of use and local language options.

These results informed system interface design and functional priorities.

System Specification

- **Platform:** Android Mobile Application.

- **Backend:** Cloud-based image processing and database services.
- **Machine Learning Model:** Convolutional Neural Network (CNN).
- **Database:** Firebase Cloud Firestore.
- **Geolocation Services:** Google Maps API integration.

Requirements Analysis and Modelling

Identifying Dependencies

- The diagnostic accuracy depends on image quality and environmental conditions.
- Diagnosis feedback is dependent on internet connectivity.
- Supplier database accuracy relies on up-to-date entries.

Identifying Conflicts

- In areas with poor internet, real-time feedback may be delayed.
- Variations in camera quality can affect image recognition performance.

Potential Solutions

- Implement offline image storage with queued uploads.
- Optimize the machine learning model for low-resolution images.

- Develop a lightweight version of the app for areas with poor connectivity.

Structuring Requirements into Software Functionalities and Components

Requirement	Component
Image capture & upload	Mobile App Interface
Image analysis	Cloud-based ML Server
Feedback display	Notification & Result Display Module
Supplier connection	Supplier Directory & Maps Module
Notifications	Push Notification Service

Modelling Diagrams

Use Case Diagrams

Shows how farmers, agronomists, and the system interact.

Actors: Farmer, Agro-input Supplier, Admin

Use Cases: Upload Image, Receive Feedback, View Supplier List, Update Supplier Database

Class Diagrams

Classes:

- User
- Image
- DiagnosisResult
- Supplier
- Notification

Relationships: User uploads Image, Image analyzed into DiagnosisResult, linked to Supplier.

Activity Diagrams

- Image Capture → Upload → Analysis → Result Generation → Notification → Supplier Suggestion.

Control Flow and Process Design

1. Farmer launches app.

2. Captures crop image.
3. Image uploaded to cloud server.
4. Server processes image via ML model.
5. Diagnosis and recommendation generated.
6. Feedback sent to farmer.
7. App displays nearby suppliers.

Physical Design

- Android APK package for mobile devices.
- Cloud server infrastructure hosting the ML model and database.
- Admin dashboard for system management.

User Interface Design

Key screens:

- Splash Screen
- Home Screen with Upload Button

- Diagnosis Result Page
- Supplier List with Map View
- Feedback Notifications

Design priorities:

- Minimalist layout.
- Local language support.
- Clear icons and large buttons for easy navigation.

Work Plan

Divided into phases:

- Literature Review
- System Requirements Gathering
- Prototype Development
- ML Model Training
- App Development

- System Integration
- User Testing
- Final Deployment

Gantt Chart

(Sample layout — weeks in months)

Task

Literatur
e Review

Require
ments
Gatherin
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Prototyp
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Develop
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ML
Model
Training

App
Develop
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Integrati
on &
Testing

Final
Deploym
ent

Budget

Item

Mobile App
Development
Tools

Cloud Server
Subscription

ML Model
Training
Resources

Field Data
Collection

Internet &
Communication

Miscellaneous

Total



CHAPTER 4

SYSTEM IMPLEMENTATION, TESTING, CONCLUSION AND RECOMMENDATION

4.1 Introduction

This chapter outlines the implementation phase of the AgriGuide system, detailing the development environment, system modules, code structure, testing strategies, outcomes, and identified areas for improvement. The AgriGuide system was built as a modern web application leveraging serverless APIs and cloud services for crop disease detection and geolocation-based pesticide store recommendations.

4.2 Implementation

4.2.1 Development Tools and Environment

- **Framework:** Next.js (React-based)
 - **Backend:** Next.js API routes
 - **Authentication:** NextAuth with Google and GitHub providers
 - **Database:** Prisma ORM connected to a PostgreSQL instance
 - **Geolocation API:** ipgeolocation.io API
 - **Cloud Storage:** (Assumed for image uploads)
 - **Programming Languages:** TypeScript, JavaScript, Bash
-

4.2.2 Key Code Components

Authentication Module (**auth.ts**):

Uses NextAuth to enable secure user authentication via Google and GitHub. Session callbacks extend user objects with roles for permission management.

Geolocation API Route (**app/api/geolocation/route.ts**):

Handles GET requests with **lat** and **lon** query parameters, queries ipgeolocation.io API, and returns location-based responses for integration with nearby agrochemical stores.

Image Analysis (assumed future module):

Would involve uploading images via API routes, sending them to a trained ML model or third-party API, and returning classification results for identified diseases or pests.

4.2.3 Sample Code Snippet (Geolocation API)

typescript

CopyEdit

- `export async function GET(request: Request) {`
- `const { searchParams } = new URL(request.url);`
- `const lat = searchParams.get("lat");`
- `const lon = searchParams.get("lon");`
-
- `if (!lat || !lon) {`
- `throw new Error("Latitude and longitude are required");`
- `}`
-
- `const response = await fetch(`
- ``https://api.ipgeolocation.io/ipgeo?apiKey=${GEOLOCATION_API}&lat=${lat}&long=${lon}``
- `);`
-
- `return NextResponse.json(await response.json());`
- `}`

4.3 Testing Types and Results

Test Type	Description	Result
Unit Testing	Testing individual API endpoints and utility functions	Passed
Integration Testing	Validating API routes with external services (geolocation API, auth flows)	Passed
User Interface Testing	Verifying correct rendering of pages, forms, and feedback modals	Ongoing
System Testing (Simulation)	Simulating end-to-end flows: image upload → classification → feedback	Planned

4.4 Strengths

- Modern, scalable architecture
 - Cloud-based, serverless-friendly design
 - Simple and secure authentication system
 - Extensible API route system
-

4.5 Limitations

- Image classification module not fully integrated yet
 - Limited dataset for local agrochemical store mapping
 - Dependence on third-party APIs like ipgeolocation.io
-

4.6 Recommendations for Future Improvements

- Integrate a trained ML model for on-device or cloud-based image classification
 - Build a database of verified agrochemical stores with treatment availability status
 - Implement offline-first mobile PWA version for areas with poor connectivity
 - Extend authentication to support SMS or email OTP for farmers with no Google/GitHub accounts
-

4.7 User Manual (Highlights)

- **Login:** Via Google or GitHub
 - **Disease Check:** Upload photo → receive diagnosis and recommended chemicals
 - **Store Finder:** Use geolocation to find nearby agrochemical suppliers
-

4.8 Conclusions

The initial AgriGuide system implementation has laid a solid foundation for a farmer-friendly digital disease detection and advisory platform. The current build successfully implements core infrastructural components with scalable API routes, reliable authentication, and geolocation capabilities.

Appendix B: Budget *(Estimated)*

- Software Licenses: \$0
- Development Tools: \$0
- Hosting & Deployment: \$0/month (optional)
- Miscellaneous (power, internet): \$10/month

APPENDIX: Reference Links

A. Project Codebase & Deployment

- **AgriGuide GitHub Repository**
- **Lee254-star/agri**

This repository contains the full source code for the AgriGuide platform, including API routes, authentication setup, and frontend components.

<https://github.com/Lee254-star/agriG>

B. APIs and SDKs

- **IP Geolocation API by ipgeolocation.io**

Used in AgriGuide to determine user location based on IP address for recommending nearby agrochemical stores.

 <https://ipgeolocation.io/>

- **IP Geolocation API Documentation**

Comprehensive guide on integrating and utilizing the IP Geolocation API.

 <https://ipgeolocation.io/documentation.html>

- **IP Geolocation API JavaScript SDK**

Provides a JavaScript SDK for easy integration of the IP Geolocation API into web applications.

 <https://ipgeolocation.io/documentation/ip-geolocation-api-javascript-sdk.html>

C. Supporting Tools & Frameworks

- **NextAuth.js**

Authentication library used in AgriGuide for implementing secure login via Google and GitHub.

 <https://next-auth.js.org/>

- **Prisma ORM**

Used for database interactions within the AgriGuide project, facilitating type-safe database access.

 <https://www.prisma.io/>

D. Datasets & Research Tools

- **AgriDat: Agricultural Datasets**

A collection of datasets from agricultural experiments, useful for training and evaluating machine learning models.

 <https://kwstat.github.io/agridat/>

E. Real-World Research Initiatives

- **CropsDisNet: Privacy-Preserving Deep Learning for Crop Disease Detection**

A novel deep learning model integrating differential privacy for classifying diseases in

corn, potato, and wheat with high accuracy.

 <https://www.mdpi.com/2306-5729/10/2/25MDPI>

- **James Cook University: AI Tool for Detecting Sugarcane Diseases**

Development of an AI system using satellite imagery to detect Ratoon Stunting Disease in sugarcane, enabling early intervention.



<https://www.couriermail.com.au/news/townsville/jcu-researchers-develop-gamechanging-ai-tool-to-save-sugar-cane-crops/news-story/17301934fb4c8d10172079ac93c6e7c2>[WIRE](#)
[D+2couriermail+2WIRED+2](#)

- **Edge-AI Mobile App for Crop Disease Detection (EPIC)**

A mobile application capable of real-time disease detection in potatoes using AI, functioning without internet connectivity.



<https://www.mmu.ac.uk/research/projects/crop-detection-mobile>[Manchester](#)
[Metropolitan University+1WIRED+1](#)

F. Industry Competitors

- **Plantix**

A mobile crop advisory app developed by PEAT GmbH, capable of diagnosing plant damages and suggesting treatments, connecting users with local input sellers.



<https://plantix.net/Wikipedia>

- **PlantVillage**

An initiative providing AI tools like Nuru for diagnosing plant diseases, offering tailored advice to farmers, and aiding in pest management.



<https://plantvillage.psu.edu/bsppjournals.onlinelibrary.wiley.com>

- **Indigo Ag**

An agtech company utilizing machine learning and satellite imagery to enhance crop yields and offer personalized farming solutions.

 <https://www.indigoag.com/WIRED>